

## Four estimates

Four versions of one problem, with **less of my working shown each time**. Item 1 is done for you; item 4 is not. **Start wherever the scaffolding stops helping you** — nobody has to announce which. The **bold labels** say what each step *accomplishes*; that is the part worth carrying forward. At each ► **Why does that step follow?**, answer in writing before going on.

|                  |                                                 |          |                                              |
|------------------|-------------------------------------------------|----------|----------------------------------------------|
| AVOGADRO         | $N_A = 6.0 \times 10^{23} \text{ mol}^{-1}$     | E. COLI  | 1 fL = $10^{-15}$ L, ~1 $\mu\text{m}$ across |
| HELA CELL        | ~20 $\mu\text{m}$ across                        | GENOME   | $4.6 \times 10^6$ bp                         |
| GFP IN CYTOPLASM | $D = 7.7 \mu\text{m}^2/\text{s}$ (Elowitz 1999) | DIVISION | 20–30 min, rich medium                       |

### 1 • Fully worked

A repressor is present at **10 nM** in *E. coli*. How many molecules is that?

Convert the concentration to moles in this volume.

$$n = cV = (10 \times 10^{-9} \text{ mol/L})(10^{-15} \text{ L}) = 10^{-23} \text{ mol}$$

Convert moles to a count.

$$N = nN_A = (10^{-23})(6.0 \times 10^{23}) = 6 \text{ molecules}$$

State the answer at the precision you actually have.

About **6 molecules per cell** — call it "of order ten". Not 6.02: the cell volume is known to maybe a factor of two, so the second digit is fiction.

**Worth carrying: 1 nM  $\approx$  1 molecule per *E. coli*.** Every other copy number in this course is that number times something.

► **Why does that step follow?** Why is it legitimate to report one significant figure when the arithmetic gave you three?

## 2 · The last step is yours

*How long does a protein take to diffuse across E. coli?*

**State the convention before the number.**

For one-dimensional displacement,  $\langle x^2 \rangle = 2Dt$ , so the time to cover a distance  $L$  is  $t \sim L^2/2D$ . (In three dimensions it is  $L^2/6D$ . The factor of three does not matter here. Not saying which one you used does.)

**Put in the measured numbers.**

$$L = 1 \mu\text{m}, \quad D = 7.7 \mu\text{m}^2/\text{s}$$

**Now you: evaluate it, and say it in words.**

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► **Why does that step follow?** Why use the *measured*  $D$  in cytoplasm rather than the value for the same protein in water — and which way would the answer go if you used the wrong one?

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### 3 · The last two steps are yours

*The same protein, in a HeLa cell instead. How long, and how many times longer than in E. coli?*

**State the convention.** Same as item 2 —  $t \sim L^2/2D$ , one-dimensional.

**Now you: everything else.** Take  $D$  to be the same, and say at the end whether that assumption makes your answer too big or too small.

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► **Why does that step follow?** The cell is 20 times wider. Why is the answer not 20 times longer?

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## 4 · Bare problem

A repressor must find **one specific operator site** in the *E. coli* genome. Suppose it searches by diffusing through the cytoplasm until it happens to collide with the right sequence.

Estimate how long that takes, and compare it to the cell cycle.

You have everything you need. State your assumptions as you go, and at the end say which one you are least confident in.

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► **The real question.** Measured association times for repressors are of order *minutes* — considerably faster than a plain three-dimensional search predicts.

Your estimate is not wrong because you made an arithmetic error. It is wrong because the *mechanism* is not what the problem said it was.

**What could the protein be doing instead?**

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*An estimate that disagrees with a measurement is not a failure. It is the only reliable way anyone has ever found a mechanism they were not looking for.*